SLIIT UNI

THE KNOWLEDGE UNIVERSITY

05A –IPv6

IE2004 Computer Networks

2026 July

Lesson outline

- IPv6 – IP protocol version 6: datagram format
- IPv6 addresses
- IPv6 auto configuration features
- IPv6 subnetting and address allocation

Key terms and concepts

- IPv6 – differences from IPv4
- IPv6 addressing
- IPv6 Autoconfiguration

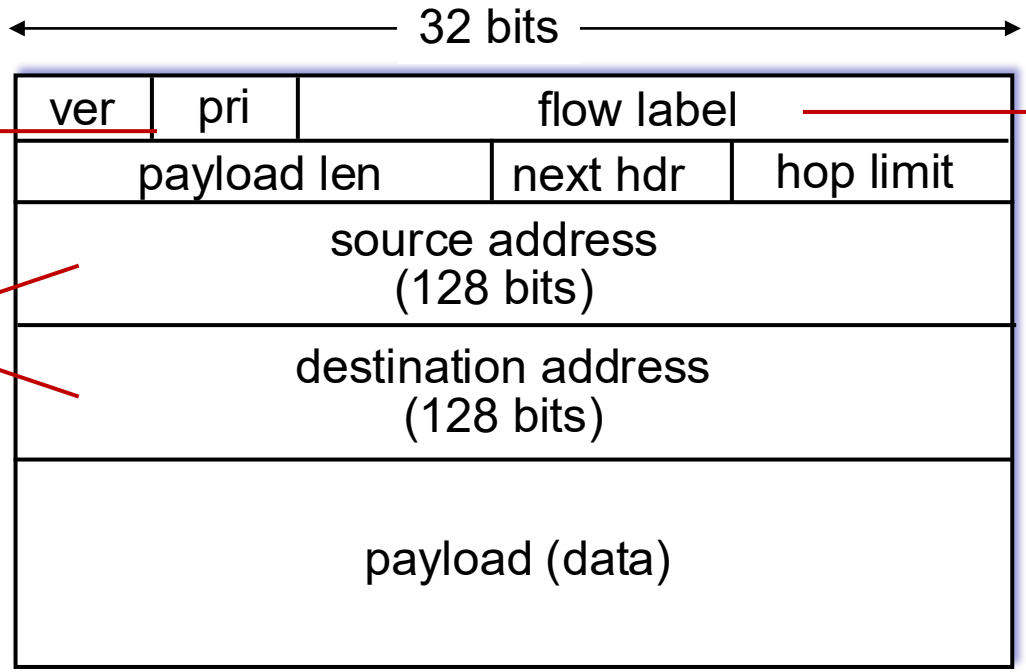
IPv6: motivation

- initial motivation: 32-bit IPv4 address space would be completely allocated
- additional motivation:
  - speed processing/forwarding: 40-byte fixed length header
  - enable different network-layer treatment of “flows”
- The initial IPv6 specification was published in 1998
  - IPv6 is not new – 26 years old in 2024 (IPv4 is 43 years old)
- Key differences between IPv4 and IPv6
  - Much larger address space (128 bits compared to 32)
  - No broadcast address – support only multicast
  - Host autoconfiguration features
  - Simpler IPv6 header (next slide)

IPv6 datagram format

priority: identify priority among datagrams in flow

128-bit IPv6 addresses

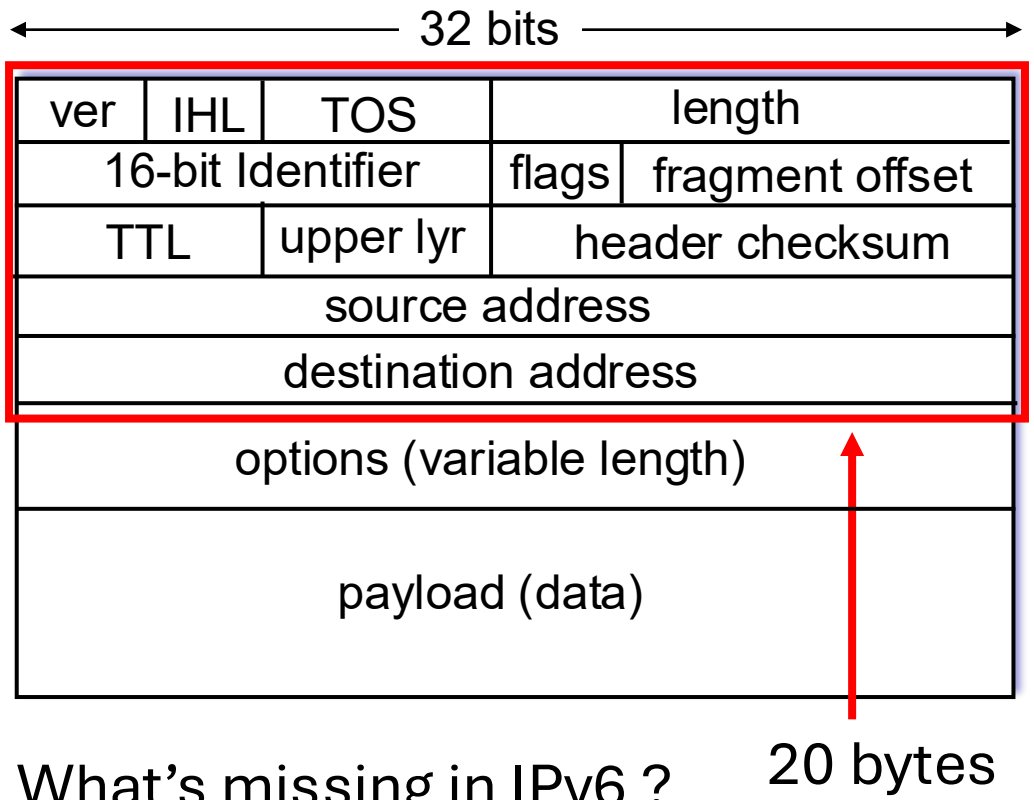


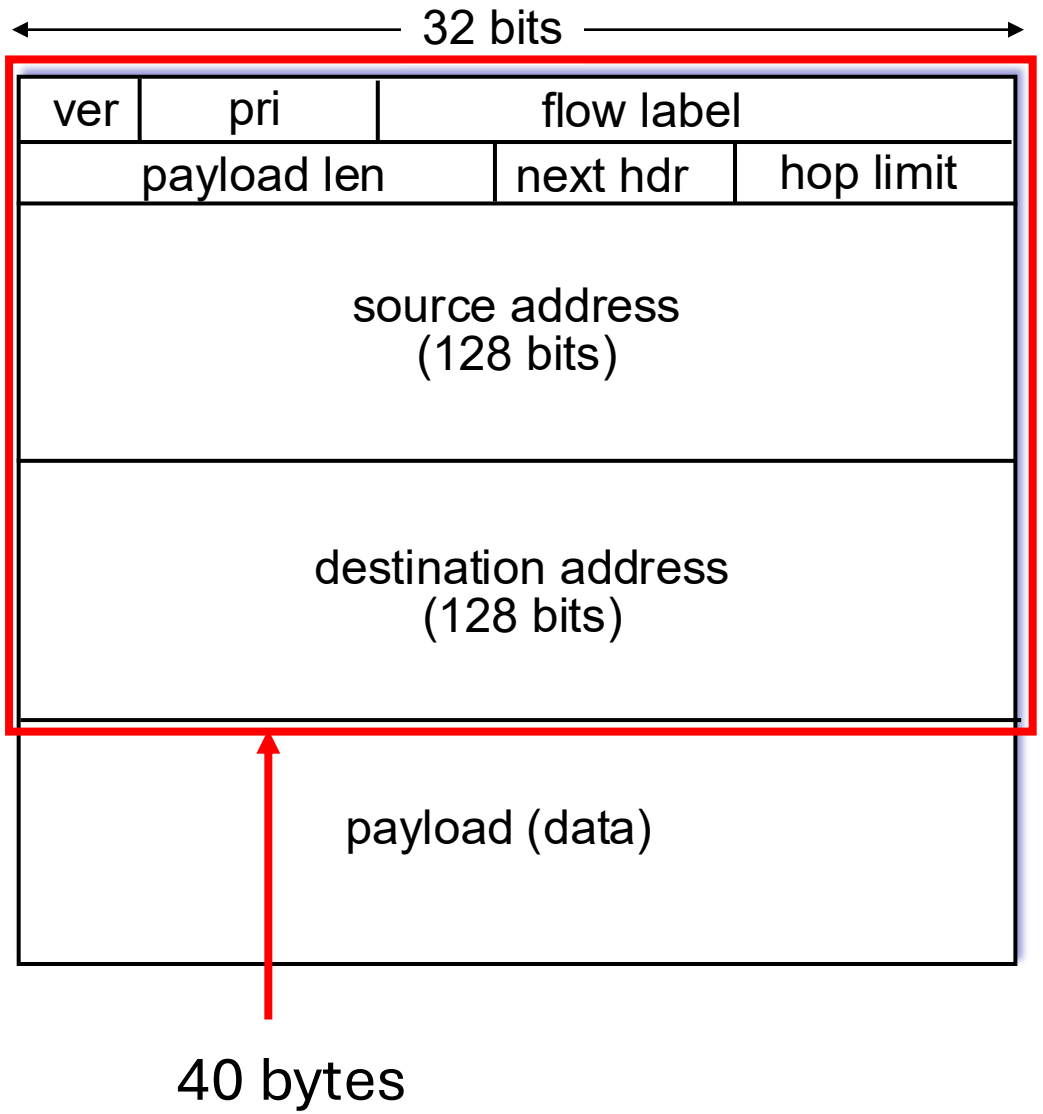
flow label: identify datagrams in same "flow." (concept of "flow" not well defined).

What's missing (compared with IPv4):

- no checksum (to speed processing at routers)
- no fragmentation/reassembly
- no options (available as upper-layer, next-header protocol at router)

IPv4 and IPv6 headers: comparison



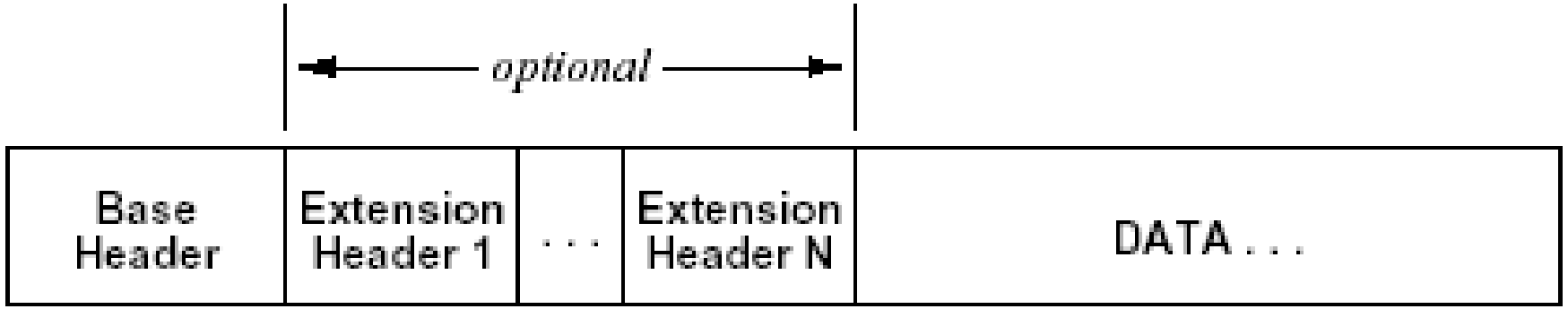


What's missing in IPv6 ?

- no checksum
- no fragmentation/reassembly
- no options

Format of IPv6 datagrams

- Instead of a variable length header, a *base header* and a variable number of *extension headers* are used
- The base header contains less information than the IPv4 header
- Options and some fields in the IPv4 header have been moved to extension headers

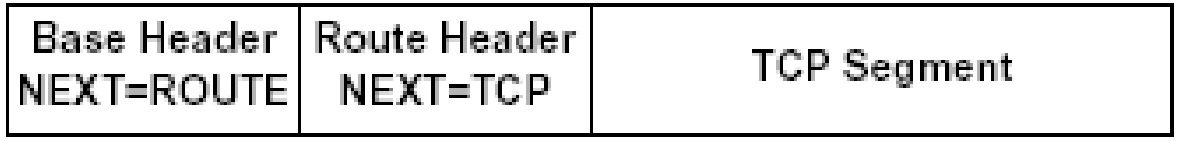


IPv6 extension headers (ctd)

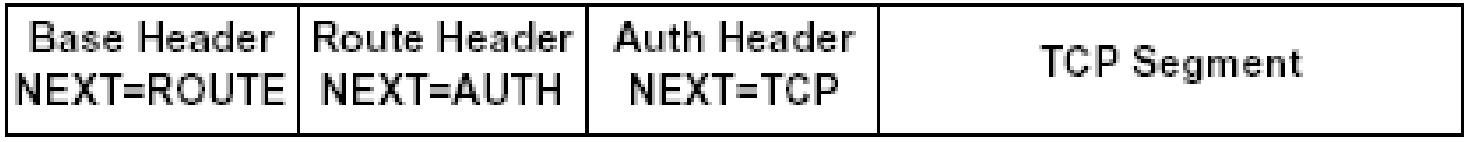
- Three IPv6 datagrams with
  a) Only the base header
  b) Base header and one extension header
  c) Base header and two extension headers



(a)



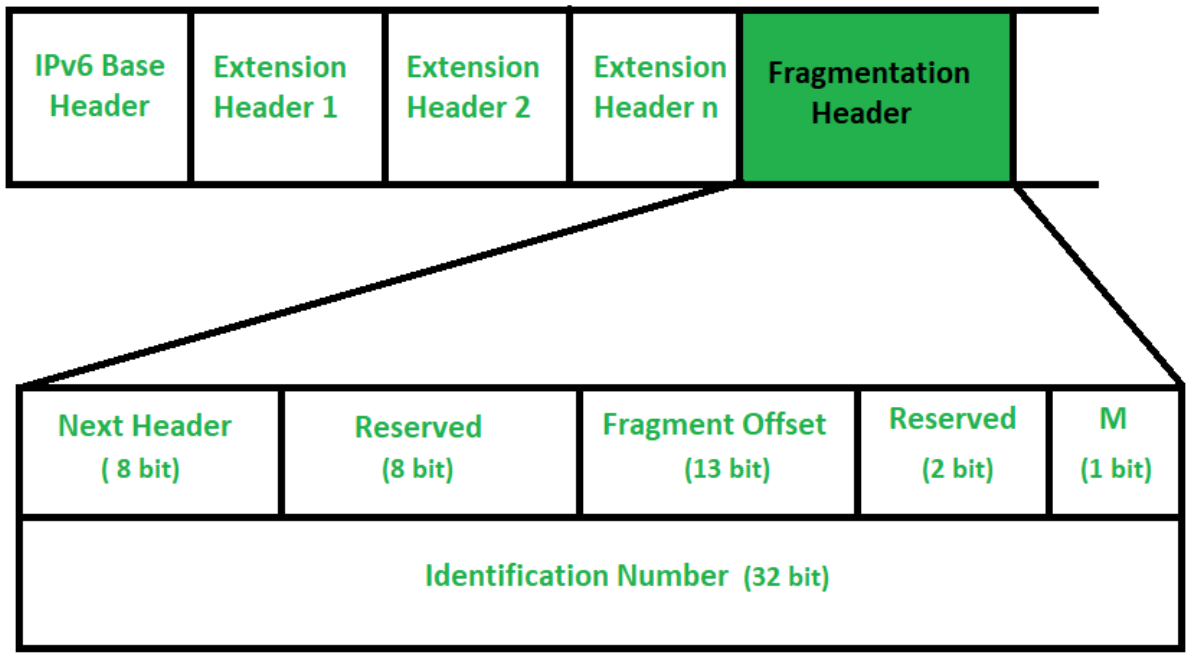
(b)



(c)

IPv6 fragmentation

- In IPv6, only the source is allowed to fragment IP datagrams
- A host can use Path MTU discovery to find out the minimum MTU along the path to a given destination
- If Path MTU discovery is not possible, host can use 1280 as MTU
- 1280 is the minimum MTU allowed with IPv6
- If fragmentation is required, Fragmentation Header (an extension header) is used



source: geeksforgeeks.org

Review questions

- Describe the fields in IPv6 datagram header and explain the function / purpose served by each field.
- For each field in IPv6 header, identify the equivalent or closely related IPv4 header field.
- What are the fields in IPv4 header that are completely missing in the IPv6 header?

Lesson outline

- IPv6 – IP protocol version 6: datagram format
- IPv6 addresses
- IPv6 auto configuration features
- IPv6 subnetting and address allocation

IPv6 Addresses

- An IPv6 address is 128 bits (16 bytes) long
- Written as 8 fields (16 bits each) separated using colons
- The colon hexadecimal notation is used to express an IPv6 address 2001:0DB8:1001:2002:3008:4800:200C:417A or 2001:0db8:1001:2002:3008:4800:200c:417a (use of lowercase recommended in RFC5952)
- Leading zero suppression
  - Leading zeros in any of the 8 fields can be omitted
  - 2001:0db8:0000:0000:0008:0800:200c:417a can be written as

    2001:db8:0:0:8:800:200c:417a
  - Only leading zeros can be suppressed. Trailing zeros cannot be.
  - So, 0800 can only be written as 800, but not as 8

IPv6 Addresses: Zero compression

- A string of repeated zeroes can be replaced by a pair of colons
  - ff05:0:0:0:0:0:0:b3 can be written as

    ff05::b3
  - 2001:db8:0:0:8:800:200c:417a can be written as

    2001:db8::8:800:200c:417a
- Zero compression can be applied only once in an address
  - 2001:0:0:0:1:0:0:2 can be written as

    2001::1:0:0:2
  - but NOT as 2001::1::2



Q: Why is this wrong?
Hint: Try to expand this to 8 fields separated by ":"

IPv6 Addresses: Zero compression (ctd)

- "::" should NOT be used to represent a single 0 field
  - 2001:db8:0:1:8:800:200c:417a

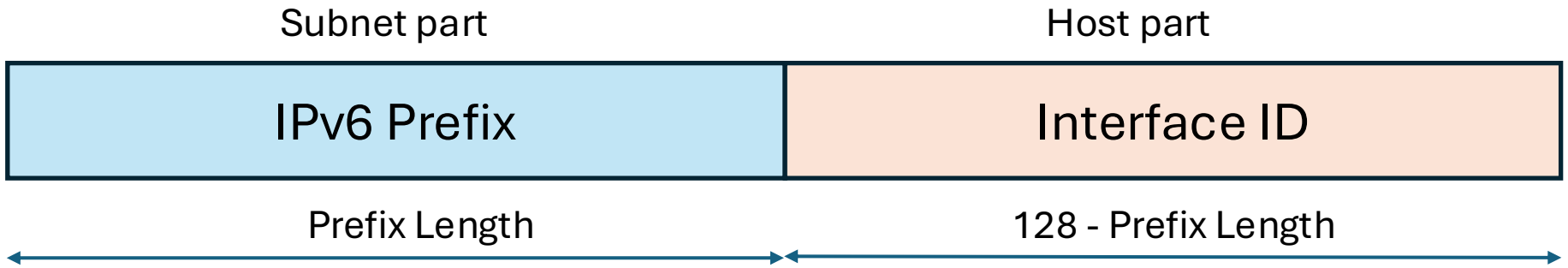
    2001:db8::1:8:800:200c:417a 
- If there are two sets of 0 fields in an address, the longest set must be represented with "::"
  - 2001:db8:0:0:1:0:0:0

    2001:db8:0:0:1::
- If there are two sets of 0 fields of equal length, the first set must be represented with "::"
  - 2001:db8:0:0:1:0:0:2

    2001:db8::1:0:0:2

IPv6 Address format

- Just like an IPv4 address, an IPv6 address has a subnet part and a host part
  - Subnet part is called the IPv6 Prefix
  - The length of the IPv6 Prefix is the Prefix Length
  - The host part is called the Host ID or Interface ID
  - The length of the host part is (128 – Prefix Length)
  - Usually prefix length is 64 bits and host part (Interface ID) is also 64 bits



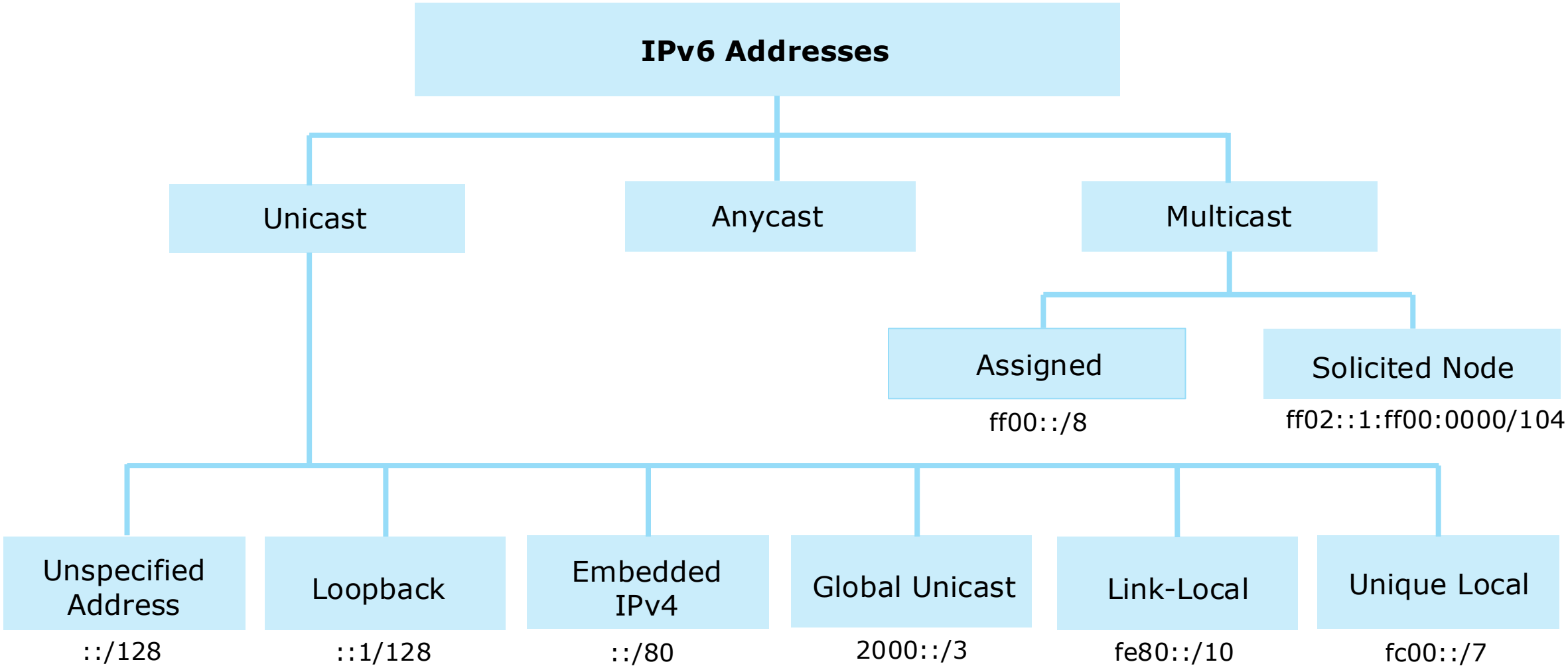
CIDR notation for IPv6 Addresses

- The CIDR notation or slash notation is used to indicate IPv6 address ranges as well as individual IPv6 addresses
  - The general format is an IPv6 Prefix or an IPv6 address followed by Prefix Length
  - If the host part is non-zero or the prefix length is 128, it indicates an address
  - Otherwise, and address space or a subnet address
- Examples of address spaces/subnets
  - 2000::/3 is the Global Unicast Address space
  - 2001:db8::/64 is a subnet in the Global Unicast Address space
  - fe80::/10 is the Link Local Address space
- Examples of addresses
  - 2001:db8::8:800:200c:417a/64
    Subnet is 2001:db8::/64 and the host ID is 8:800:200c:417a
  - ::/128 is the unspecified address
  - ::1/128 is the loopback address

IPv6 Address types

- There are three address types in IPv6
- Unicast addresses
  - Destination address specifies a single computer
  - The packet must be routed to the destination via a shortest path
- Anycast addresses
  - Destination address specifies a set of computers, possibly at different locations. All share a single address
  - Packet must be routed to any one in the set along a shortest path
- Multicast addresses
  - Destination address specifies a set of computers, possibly at multiple locations
  - One copy of the packet should be sent to each member of the group using hardware multicast or broadcast, if applicable

IPv6 Address types

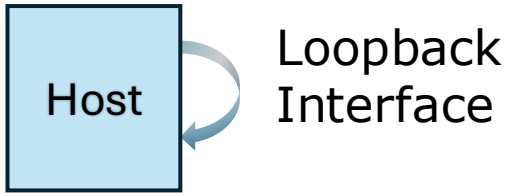


Broadcast and Multicast

- Recall: IPv4 provides a broadcast mechanism
  - all 1's to broadcast to all hosts on the local network (limited broadcast)
  - network number followed by all 1's to broadcast on a particular network (directed broadcast)
- But in IPv6 there is no broadcast address for a network
- The link-local multicast address ff02::1 (all nodes multicast address)
  - Targets all nodes on the local link (i.e., all hosts on the local network)
  - A packet destined to ff02::1 is received by all IPv6-enabled hosts on local network
  - Serves the purpose of an IPv4-like broadcast within the local network
  - But the effect of broadcast is achieved through multicast, and is limited to local network
- So, limited broadcast is possible with IPv6 multicast address ff02::1
- Directed broadcast is not possible with IPv6

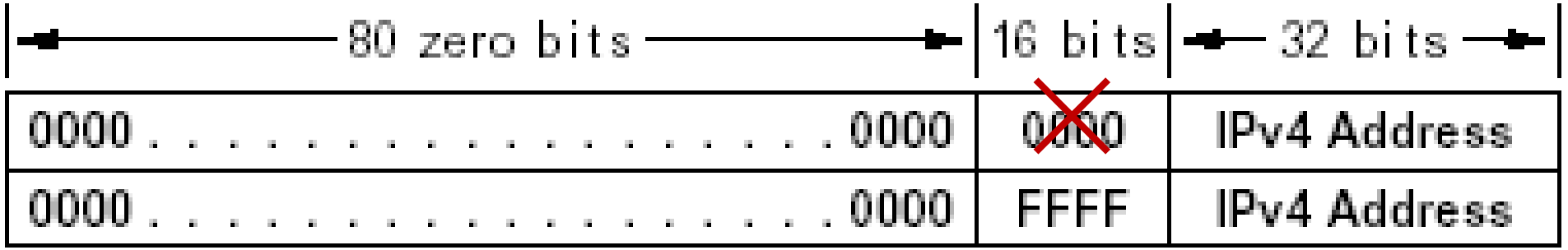
Unspecified and Loopback addresses

- The address of all 0s - 0:0:0:0:0:0:0:0/128 (or ::/128) is the unspecified address which hosts may use as the source address when they are booting
  - The unspecified address may not be used in the destination address field
  - This is like 0.0.0.0/32 for IPv4
- The loopback address is 0:0:0:0:0:0:0:1/128 (or ::1/128)
  - A packet destined to the loopback address must not leave the local machine
  - This is like 127.0.0.1/32 for IPv4



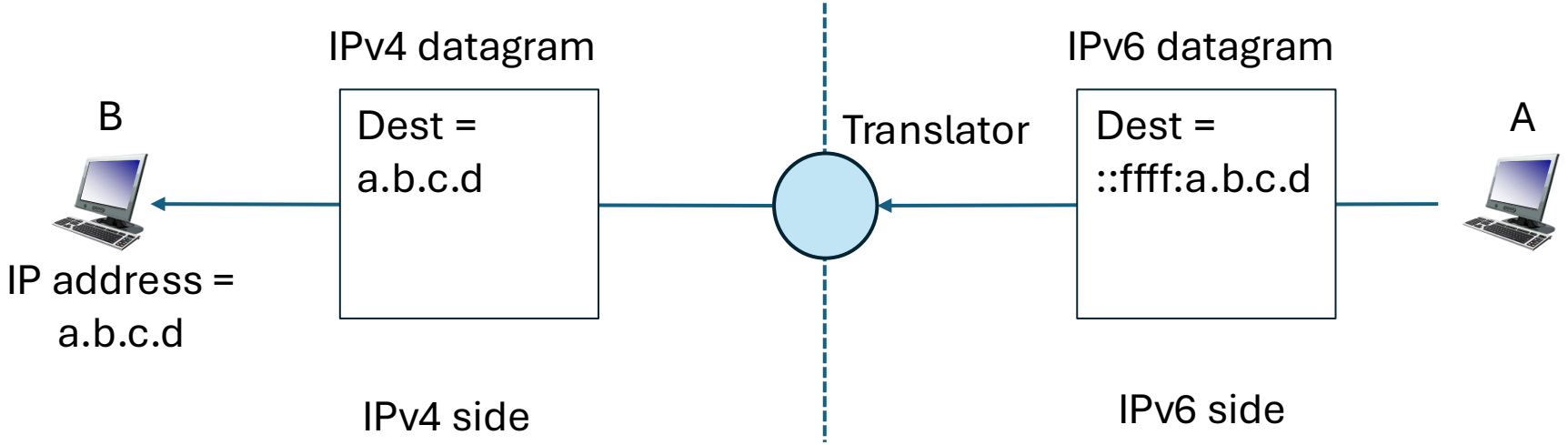
Embedded IPv4 Addresses

- An IPv4 address can be embedded in an IPv6 address
  - Previously, 80 zeros followed by 0000 or ffff and the 32-bit IPv4 address
  - But using 0000 is now deprecated (by RFC4291)
- Example
  - Consider IPv4 address 223.1.1.1
  - It can be represented as ::ffff:223.1.1.1 (this is mixed notation)
  - In pure IPv6 notation it is ::ffff:df01:101



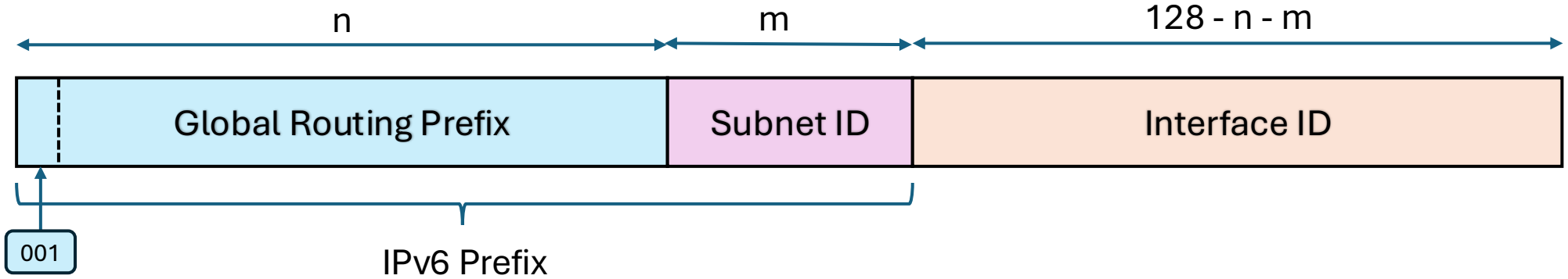
Using embedded IPv4 Addresses

- A translator can be used for communication between IPv4 and IPv6 hosts
  - The translator uses pure IPv4 to communicate with the IPv4 host, and pure IPv6 to communicate with the IPv6 host
  - The IPv6 host A can send an IPv6 packet with the embedded IPv4 address of host B to the translator
  - The translator will translate this to an IPv4 packet with B's IPv4 address
  - Similar translation needed for packets going from IPv4 side to IPv6 side



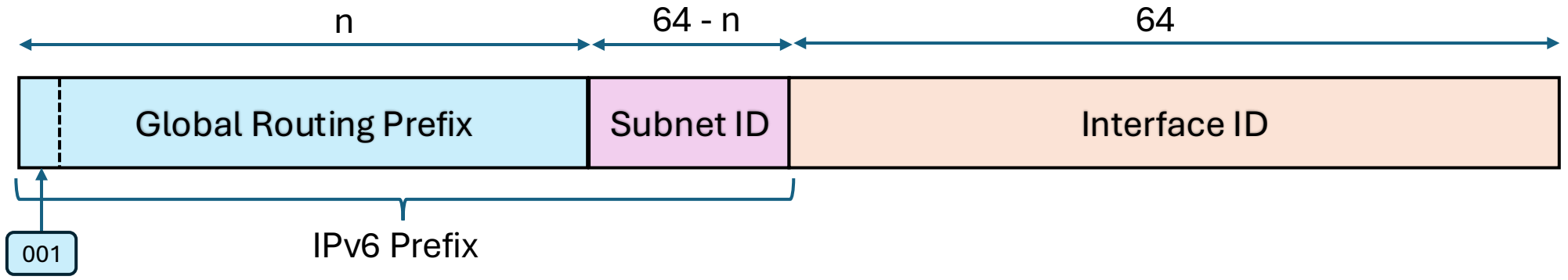
Global Unicast Addresses

- IPv6 global unicast addresses (GUAs) are globally unique and routable on the IPv6 internet
  - Currently, only GUAs starting with 001 (2000::/3) are being assigned
  - Begins with a hexadecimal 2 or a 3
- The format of a GUA Address is as shown below
  - IPv6 Prefix part is broken into two fields: Global Routing Prefix + Subnet ID
  - Global Routing Prefix: the prefix, defines the address range provided by an address provider (e.g. ISP) to a customer or site
  - Subnet ID: Used by an organization to identify subnets within the site



Global Unicast Addresses (ctd)

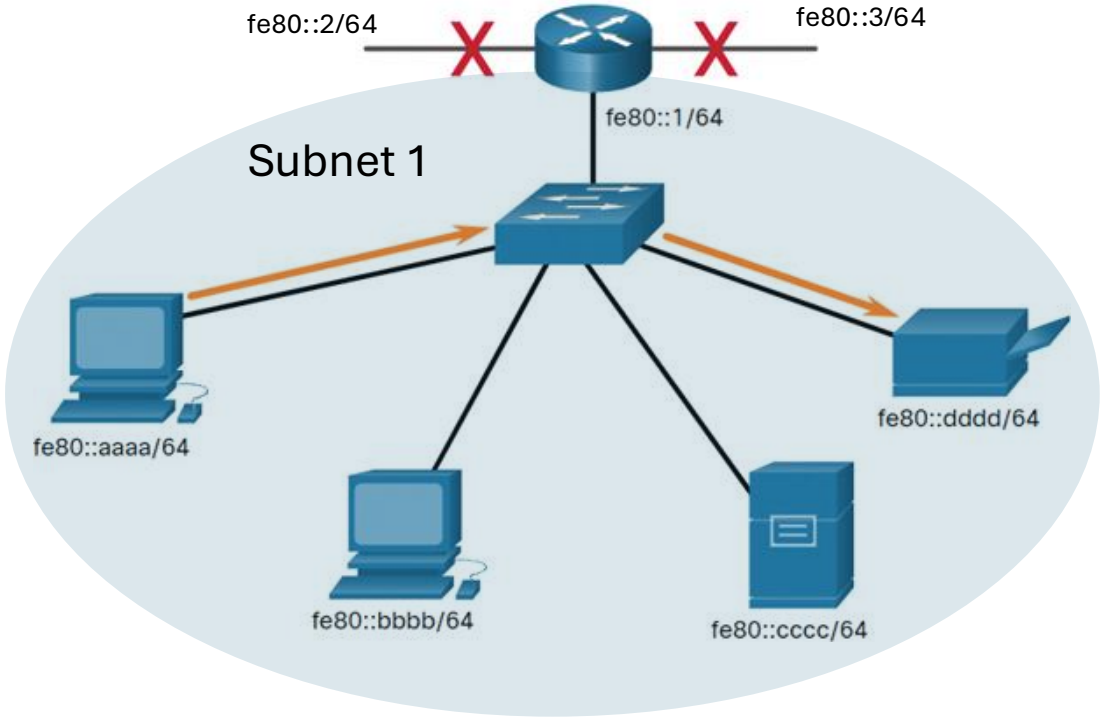
- The number of bits allocated for the Interface ID is usually 64 bits
- 64 bits for Global Routing Prefix + Subnet ID
  - IPv6 Prefix range is 2000::/64 to 3fff::/64
- Usually, a site is allocated a /48 address space (n=48)
  - Allows 16 bits for the Subnet ID (2^{16} = 65536 subnets per site)
- ISPs usually get at least a /32 address space from which they can delegate /48 address spaces for customer sites

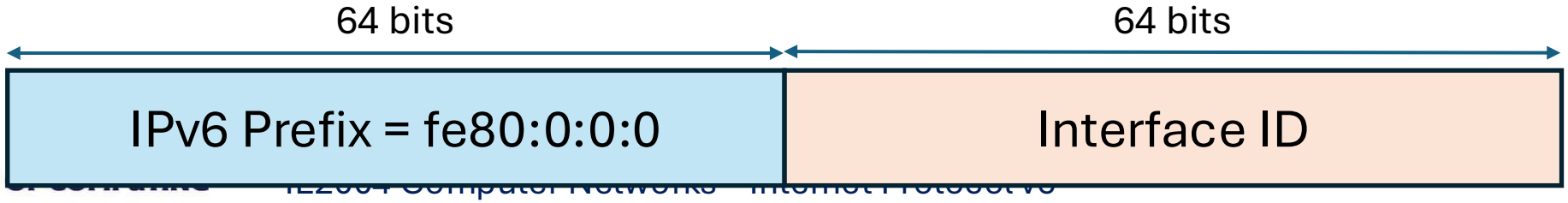


Link Local Addresses (LLA)

- An IPv6 link-local address (LLA) enables a device to communicate with other IPv6-enabled devices on the same link (subnet) and only on that link (subnet)
- Every IPv6-enabled network interface must have an LLA
  - A device with multiple network interfaces should have an LLA for each interface
- fe80::/10 space is reserved for LLAs
  - But in practice, LLAs use fe80::/64
- A packet with a source or destination LLA cannot be routed beyond the link (subnet) on which it originated
  - Hosts on Subnet 1 cannot use LLAs to reach the router interfaces marked in red



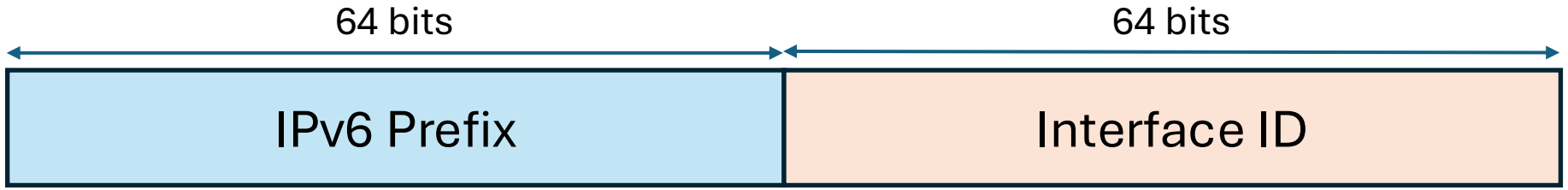




Interface ID

- The low order 64 bits of an IPv6 unicast address is the Interface ID
  - Applicable to GUAs, LLAs and other unicast address types
- Interface ID is used to identify interfaces on a link (subnet)
  - They are required to be unique within a subnet
- Can be assigned in different ways:
  - Manually configured
  - Automatically generated
    - Modified EUI-64 mechanism
    - Pseudo-random number

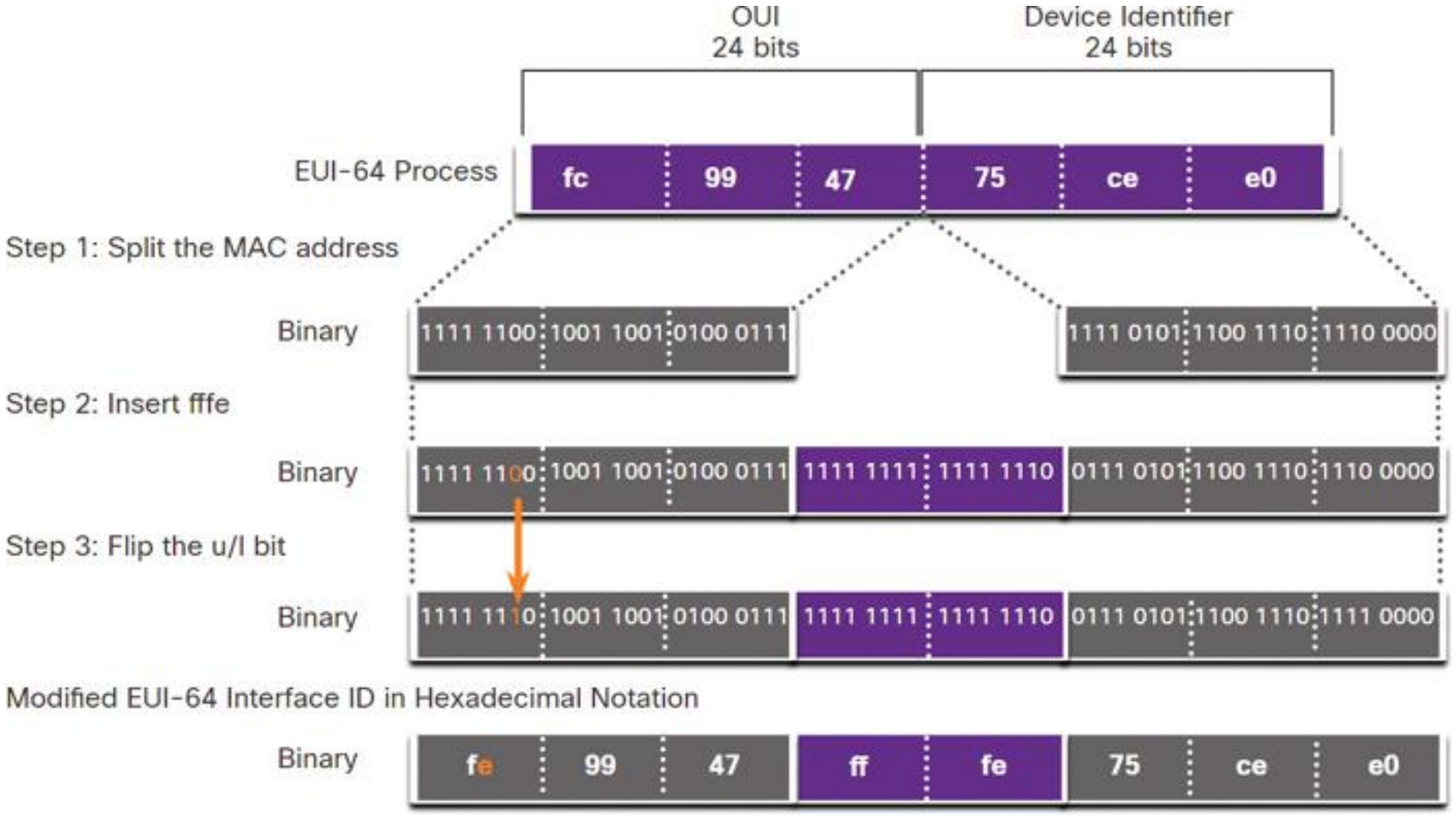
If the Interface ID is automatically generated, its uniqueness within the subnet must be verified using Duplicate Address Detection (DAD)



Modified EUI-64 mechanism

- IEEE EUI-64 process defines a standard format for globally unique 64-bit Interface IDs
  - Uses the MAC (L2) address of the interface to generate the Interface ID
- Ethernet MAC addresses are 48 bits long, and are made up of two parts:
  - Organizationally unique identifier (OUI): A 24-bit vendor code assigned by IEEE
  - Device identifier: A unique 24-bit value within a common OUI
- To form an Interface ID
  - Insert fffe (16 bits) between OUI and Device Identifier of the MAC address
  - Invert the 7th bit (the universal/local [U/L] bit)
- For example, if the MAC address is fc:99:47:75:ce:e0
  - Inserting fffe gives fc99:47ff:fe75:cee0
  - Inverting 7th bit gives fe99:47ff:fe75:cee0

Modified EUI-64 mechanism (ctd)

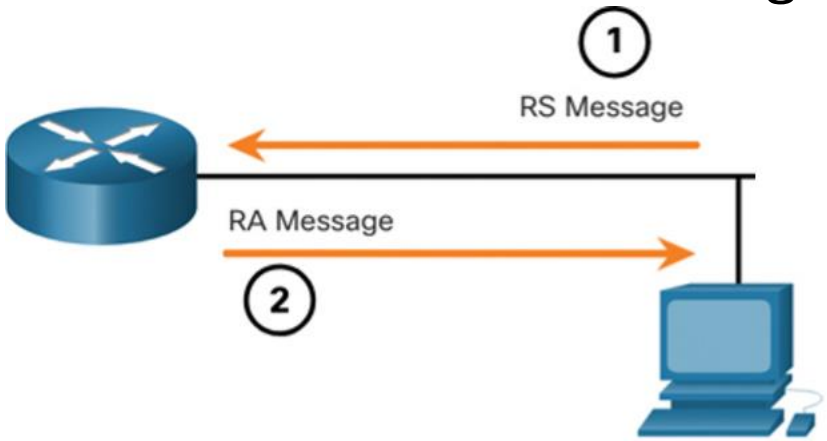


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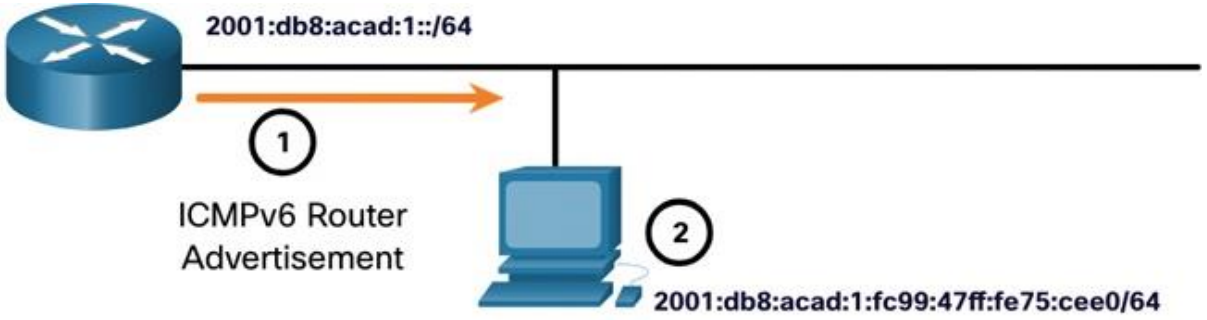
Autoconfiguration

- Known as Stateless Address Autoconfiguration or SLAAC
- Initially a host is permitted to use only an LLA for an interface
- The host then sends a router solicitation (RS) message, requesting a GUA Prefix that it can use
- A router on the local network responds with a router advertisement (RA) message providing valid GUA Prefixes that the host may use
  - Routers send RA messages periodically, or in response to a RS message



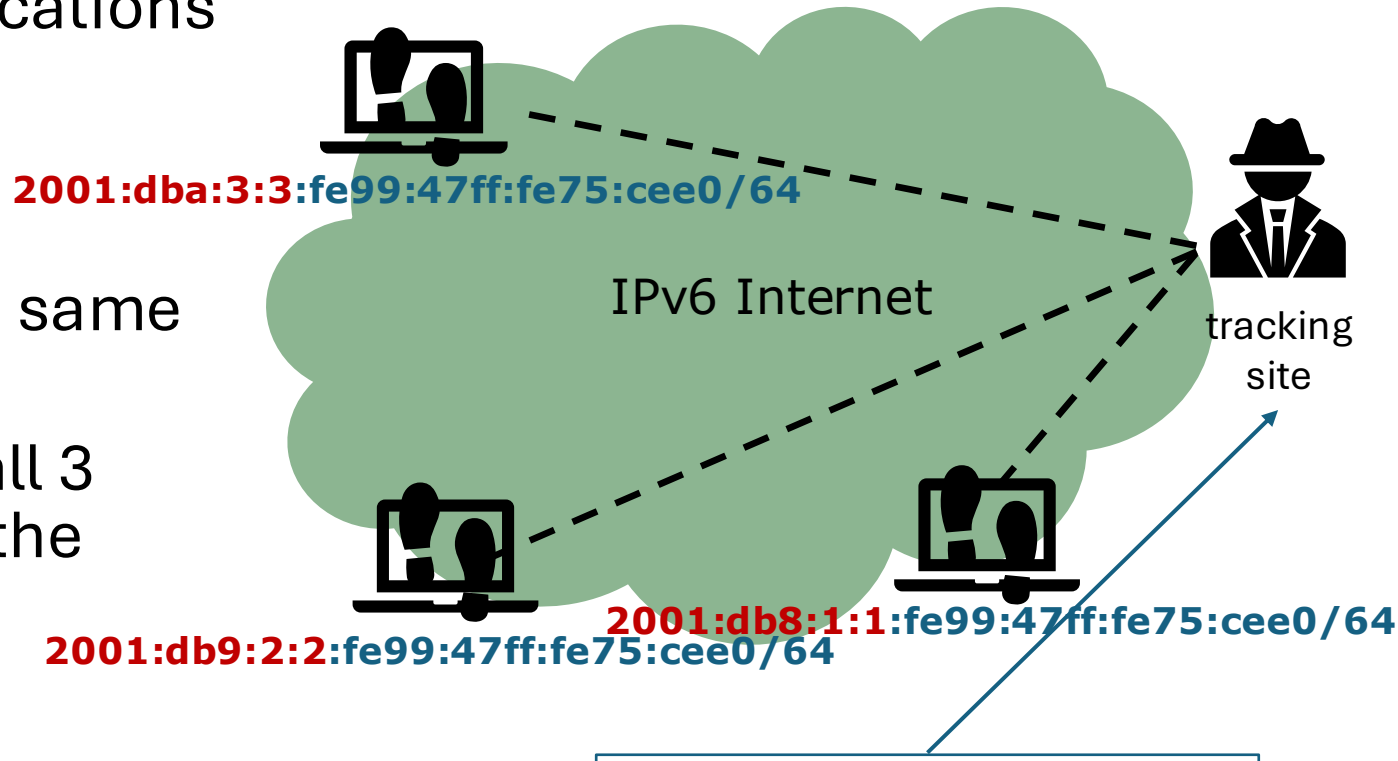
Autoconfiguration (ctd)

- The GUA of the host is formed as GUA Prefix + Interface ID
- The host can now use both the GUA and the LLA
  - LLA can only be used for communications within the subnet
  - GUA can be used for all communications (including within subnet)
- The prefixes are valid for a finite duration, specified in the advertisement
  - After the expiry of the duration, the host must repeat the process



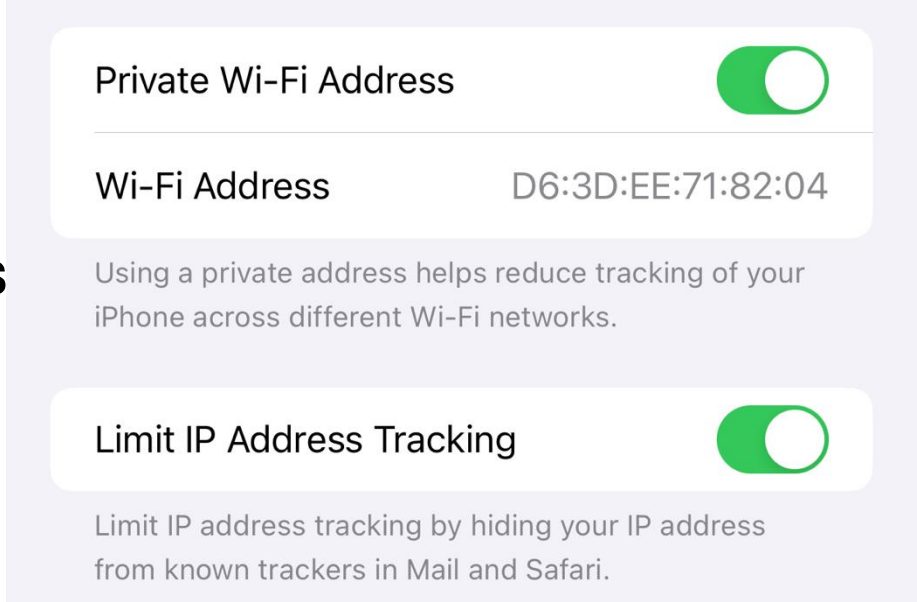
Security concerns related to use of EUI-64

- Consider a mobile device connecting to the Internet from different locations
  - Office during daytime
  - Home outside working hours
  - Hotel when traveling
- The interface ID will remain the same for all three connections
- A site the user accesses from all 3 locations can identify that it is the same device connecting
- User privacy lost
  - Can track user's movements
  - Can target the user in a malicious way
  - Directed advertising!



Pseudo-Random Interface ID

- As a solution to privacy issues associated with EUI-64, hosts can use a randomly generated Interface ID every time it uses autoconfiguration
- But for devices like servers which accept incoming connections, the Interface ID must not change
- A common practice is to use a stable address and temporary addresses
  - Stable address: Use EUI-64 generated Interface ID for accepting incoming connections
  - Temporary address: Use a pseudo-random generated Interface ID for outgoing connections
- Most operating systems now provide options to use a fixed MAC address or a random MAC address
  - Using EUI-64 with a randomly generated MAC address will give different Interface IDs



IPv6 configuration: Exercise

1. Connect to the OpenLMS Wi-Fi network
2. Now open a browser on your device, and go to https://whatismyipaddress.com/
3. Note down your public IPv6 address and the IPv4 address
4. Find the IPv4 and IPv6 addresses of your device wireless interface
   - On iPhone devices, go to Settings -> Wi-Fi and then tap on ⓘ icon next to OpenLMS
   - On Android devices, go to Settings -> Wi-Fi and Network Connections and then tap on the OpenLMS network name
   - On notebooks go to Control Panel / System settings and see properties of the wireless network interface
5. Note that NAT is used for IPv4 but not for IPv6
   - IPv4 interface address and address shown by whatismyipaddress.com are different
   - IPv6 interface address and address shown by whatismyipaddress.com are the same

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IPv6 Subnetting

- Subnetting in IPv6 is a lot simpler than IPv4 subnetting
  - All the subnets are /64 or larger
  - In IPv6, subnetting is usually done at a nibble boundary
  - Nibble = Half of a byte (4 bits or 1 hexadecimal digit)
  - Subnet sizes are /64, /60, /56 or /52
- What we need to do is divide the address space we have
  - Usually a /48 address space is available for an organization
  - Depending on the organization we can allocate portions of this space to sites, departments, buildings, floors etc
  - But each single IPv6 network is /64 in size
  - So we have 16 bits (64 – 48 = 16) to encode parameters like sites, departments, buildings, floors etc

IPv6 Subnetting example

- Let's say we have the IPv6 space 2001:db8:1::/48 available to us
- We can allocate subnets according to the following scheme
  - We use the subnet numbering scheme 2001:db8:1:WXYZ::/64
  - The first nibble W can indicate the site (1 – main, 2- remote site A, etc)
  - The second nibble can indicate the building number (1 – Building A, 2- Building B, …)
  - The third nibble can indicate the department (1-admin, 2 – HR,… etc)
  - The fourth nibble can encode other relevant details like user groups (1 – Staff, 2- Students, … etc)
- The resulting subnetting would be
  - Usually reserve 2001:db8:1::/56 for the network core
  - 2001:db8:1:1111::/64 would be main site, Building A, Admin department, staff user subnet
  - 2001:db8:1:2121::/64 would be remote site A, Building A, HR department, staff user subnet

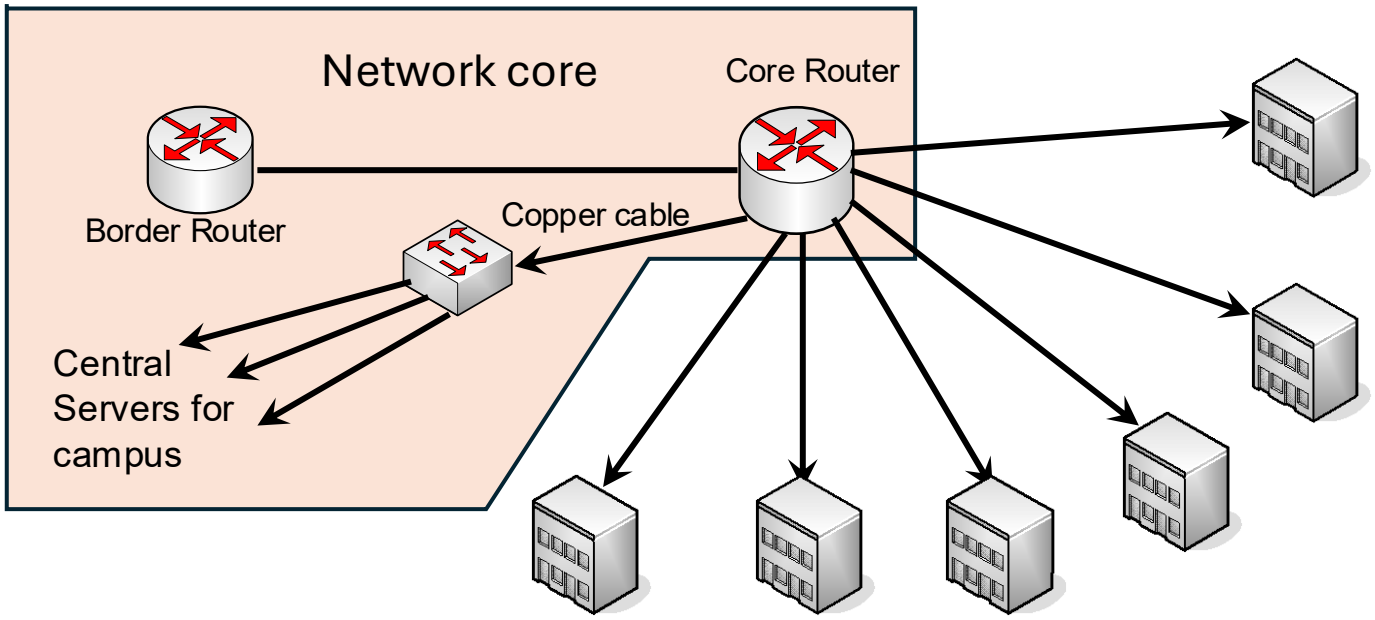
IPv6 Subnetting example 2

- Another possible scheme, suitable for a university is 2001:db8:1:XYYZ::/64 where
  - X indicates the faculty (0 – IT Services, 1 to15 – other faculties)
  - YY indicates the department (256 departments per faculty)
  - Z indicates the LAN (16 LANs per department: 0-wired, 1-wireless, …)
- The resulting subnetting would be
  - Usually reserve 2001:db8:1::/56 for the network core
  - 2001:db8:1:1010::/64 would be faculty 1, department 1, wired subnet
  - 2001:db8:1:2101::/64 would be faculty 2, department 16, wireless subnet
  - …..

IPv6 Subnetting / address allocation exercise

- A campus network has several buildings and a campus wide wireless network. The IPv6 address space 2001:db8:1::/48 is available for use in the campus. Reserve the first /56 of this address space for the network core and allocate IPv6 address ranges for the subnets indicated below.

Network	Hosts
Science Building	120
Arts Building	50
Engineering Building	200
Library	80
Administration Building	40
Staff & Student Hostel	60
Wireless Network	350



References

- Chapter 4 - The Network Layer: Data Plane
  James F. Kurose and Keith W. Ross, Computer Networking – A Top-Down Approach, (8th Edition), Pearson, 2020
- Chapter 4 - IPv6 Address Format and Address Types
  Graziani, R., IPv6 Fundamentals: A Straightforward Approach to Understanding IPv6 (2nd Edition), Cisco Press, 2017

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